Longest Common Substring with At Most 1 Mismatch

HackerRank

You need to find the length of the longest pair of substrings (one from A and one from B) that have the exact same length and differ in at most 1 position.

Additionally, if the two substrings differ in exactly 1 position, the two mismatched characters must either both be vowels ('a', 'e', 'i', 'o', 'u') or both be consonants.

If the mismatched characters do not share the same vowel/consonant type, the mismatch is considered invalid and the substrings do not meet the criteria.

Find the length of the longest valid pair of substrings.

Input Format

The first line contains a string, A, denoting the first string.

The second line contains a string, B, denoting the second string.

Constraints

1 <= |A| <= 10^3

1 <= |B| <= 10^3

Output Format

The output contains an integer value

Sample Input 0

```
hello
hallo
```

Sample Output 0

```
5
```

Explanation 0

The two strings differ at index 1. In string A, the character is 'e', and in string B, it is 'a'. Both 'e' and 'a' are vowels, which makes this a valid mismatch. Since there is only 1 valid mismatch in the entire length of the strings, the longest valid pair of substrings is the entire strings themselves. The length is 5.

Sample Input 1

```
hello
hxllo
```

Sample Output 1

```
3
```

Explanation 1

The two strings differ at index 1. In string A, the character is 'e' (a vowel), and in string B, it is 'x' (a consonant). Because they are not of the same type, this mismatch is invalid. The longest valid matching substrings without invalid mismatches are "llo" (length 3) and "h" (length 1). The maximum length is 3.

Sample Input 2

```
abcdef
abzdef
```

Sample Output 2

```
6
```

Explanation 2

The strings differ at index 2 ('c' in A and 'z' in B). Both 'c' and 'z' are consonants, so the mismatch is valid. The entire strings can be matched together with exactly 1 valid mismatch. The length is 6.